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Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – December 2025

First Semester

Artificial Intelligence & Data Sciences

AI Optimization Techniques for Healthcare Resource Management

Duration: Two hours

Maximum: 50 Marks

**Course Outcomes (COs):**

| CO | Course Outcomes |
|---|---|
| CO01 | Understand healthcare systems, resources, and challenges, and formulate optimization problems. |
| CO02 | Apply classical optimization techniques (LP, IP) and AI/ML models for hospital resource management. |
| CO03 | Implement metaheuristic and evolutionary algorithms for healthcare scheduling, allocation, and routing. |
| CO04 | Analyze and design AI-based decision support systems for patient flow, bed management, and emergency response. |

**Answer all questions**

1) Formulate a linear programming problem for optimizing bed allotment for amrita hospital. [5 marks] [CO02][BTL3 Level]
2) A startup builds an AI tool that detects cardiac arrhythmia using smartwatches. Discuss approval pathway (e.g., US FDA/India), risks, and clinical trials. [5 marks] [CO01][BTL2 Level]
3) Propose an AI-enabled non-invasive neonatal monitoring architecture integrating respiration, $SpO_2$, heart rate, and jaundice level. [5 marks] [CO01] [BTL3 Level]
4) Describe the process of predictive modeling for forecasting equipment and drug demand in a hospital. [5 marks] [CO01][BTL2 Level]
5) A hospital has limited ventilators, doctors, and oxygen supply. Formulate an LPP to optimize emergency case handling. [5 marks] [CO02] [BT4Level]
6) Define optimization in healthcare resource management and explain its importance. [5 marks] [CO02] [BT2Level]



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7) A multi-specialty hospital has two diagnostic laboratories that operate 7 hours per day each. The hospital performs three types of diagnostic procedures:
MRI Scan: requires 75 minutes and yields ₹6,000 revenue
CT Scan: requires 45 minutes and yields ₹4,000 revenue
Ultrasound: requires 20 minutes and yields ₹1,500 revenue
Due to staffing and patient-load limits, a maximum of 5 MRI, 8 CT scans, and 12 Ultrasound procedures can be performed per day. Task:
   a) Formulate a Linear Programming Problem to maximize daily revenue.
   b) Solve the LPP using graphical or simplex method and compute the optimal number of MRI, CT, and Ultrasound procedures to be scheduled. [8 marks] [CO02] [BT5Level]

8) A hospital needs to allocate 100 ambulances across two departments: General, ICU, to optimize patient waiting time while maximizing utilization.
   a) Formulate this as an optimization problem.
   b) Suggest whether linear or integer programming is suitable and justify your choice.
   c) Discuss how AI could further enhance this allocation process. [8 Marks] [CO01] [BTL5]

9) Differentiate between supervised and unsupervised learning by using a CT scan X-ray Pneumonia problem. [4 Marks] [CO01] [BTL2]

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**Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table**

| CO | Marks | BTL | Marks |
|---|---|---|---|
| CO01 | **27** | BTL 1 | |
| CO02 | **23** | BTL 2 | **19** |
| CO03 | - | BTL 3 | **10** |
| CO04 | - | BTL 4 | **5** |
| CO05 | - | BTL 5 | **16** |
| CO06 | - | BTL 6 | - |

**Bloom's Taxonomy Levels (attached for reference)**

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply



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